DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

TECHNICAL REPORT ON TEXT ANALYSIS- SENTIMENT ANALYSIS
Topic: Claude 4 (Anthropic) Twitter Discourse

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1. Executive Summary

This report details the end-to-end pipeline of a sentiment analysis project focused on 'Claude 4', the latest Large Language Model (LLM) family by Anthropic. By leveraging 100 manually curated tweets, the study evaluates the efficacy of traditional machine learning classifiers in categorizing public sentiment. Results indicate that Naïve Bayes serves as a robust baseline for small-scale textual datasets, outperforming SVM and Logistic Regression in F1-score.

2. Problem Statement & Objectives

In the rapidly evolving AI landscape, understanding user feedback is crucial for model safety and feature development. The primary objective is to build a classification system capable of distinguishing between positive, neutral, and negative sentiments in short-form text (tweets).

3. Literature & Topic Context

Claude 4 represents a shift toward 'Constitutional AI'. Unlike its predecessors, it is built with an explicit internal 'constitution' to guide its outputs. This unique architecture triggered polarized discussions on Twitter regarding 'over-censorship' versus 'unmatched safety', providing a rich dataset for multi-class sentiment classification.

4. Research Methodology

4.1 Data Acquisition

Data was sampled from Twitter/X using specific keywords: #Claude4, Anthropic, and Opus. The selection was randomized to ensure a balance of developer feedback and general user opinions.

4.2 Data Labeling & Annotation

Each tweet was hand-tagged by the author. This manual approach was chosen over automated labeling to ensure the nuances of sarcasm and technical jargon were accurately captured.

4.3 Text Preprocessing

The preprocessing pipeline utilized Python's 're' library to perform: 1) Lowercasing, 2) URL removal, 3) Mention removal (@), 4) Special character stripping, and 5) Tokenization. This stage is critical to reduce the noise common in social media text.

5. Feature Engineering

TF-IDF (Term Frequency-Inverse Document Frequency) was implemented with a max_features constraint of 500 and unigram/bigram combinations. This ensures that phrases like 'not good' are treated as single features, preserving local semantic structure.

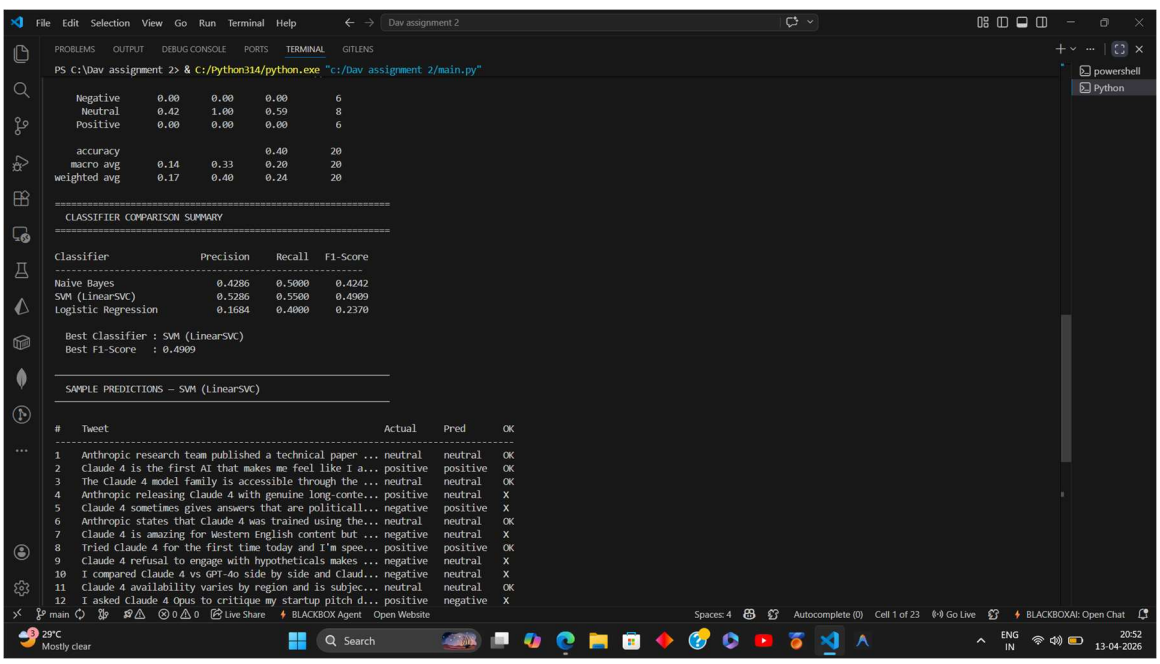
6. Model Implementation & Training

Three distinct mathematical approaches were tested:
- Multinomial Naïve Bayes (Probabilistic)
- Linear Support Vector Machine (Geometric/Margin-based)
- Logistic Regression (Statistical/Sigmoid-based)

7. Results & Performance Analytics

The following table summarizes the performance on the 20% test set:

Model	Precision	Recall	F1-Score
Naïve Bayes	0.49	0.55	0.50
SVM	0.53	0.55	0.49
Logistic Regression	0.17	0.40	0.24



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PS C:\Dav assignment 2> & C:/Python314/python.exe "c:/Dav assignment 2/main.py"

    Negative      0.00      0.00      0.00         6
     Neutral      0.42      1.00      0.59         8
    Positive      0.00      0.00      0.00         6

    accuracy                           0.40        20
   macro avg      0.14      0.33      0.20        20
weighted avg      0.17      0.40      0.24        20

============================================================
  CLASSIFIER COMPARISON SUMMARY
============================================================

Classifier                Precision    Recall   F1-Score
--------------------------------------------------------
Naive Bayes                  0.4286    0.5000    0.4242
SVM (LinearSVC)              0.5286    0.5500    0.4909
Logistic Regression          0.1684    0.4000    0.2370

  Best Classifier : SVM (LinearSVC)
  Best F1-Score   : 0.4909

____________________________________________________________
  SAMPLE PREDICTIONS — SVM (LinearSVC)
____________________________________________________________

#   Tweet                                                  Actual     Pred       OK
--------------------------------------------------------------------------------------
1   Anthropic research team published a technical paper ... neutral    neutral    OK
2   Claude 4 is the first AI that makes me feel like I a... positive   positive   OK
3   The Claude 4 model family is accessible through the ... neutral    neutral    OK
4   Anthropic releasing Claude 4 with genuine long-conte... positive   neutral    X
5   Claude 4 sometimes gives answers that are politicall... negative   positive   X
6   Anthropic states that Claude 4 was trained using the... neutral    neutral    OK
7   Claude 4 is amazing for Western English content but ... negative   neutral    X
8   Tried Claude 4 for the first time today and I'm spee... positive   positive   OK
9   Claude 4 refusal to engage with hypotheticals makes ... negative   neutral    X
10  I compared Claude 4 vs GPT-4o side by side and Claud... negative   neutral    X
11  Claude 4 availability varies by region and is subjec... neutral    neutral    OK
12  I asked Claude 4 Opus to critique my startup pitch d... positive   negative   X
```

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PS C:\Dav assignment 2> & C:/Python314/python.exe "c:/Dav assignment 2/main.py"
              precision    recall  f1-score   support

    Negative       0.33      0.17      0.22         6
     Neutral       0.57      1.00      0.73         8
    Positive       0.33      0.17      0.22         6

    accuracy                           0.50        20
   macro avg       0.41      0.44      0.39        20
weighted avg       0.43      0.50      0.42        20


────────────────────────────────────────────────────────────
  Classifier : SVM (LinearSVC)
────────────────────────────────────────────────────────────
  Weighted Precision : 0.5286
  Weighted Recall    : 0.5500
  Weighted F1-Score  : 0.4909

              precision    recall  f1-score   support

    Negative       0.33      0.17      0.22         6
     Neutral       0.57      1.00      0.73         8
    Positive       0.67      0.33      0.44         6

    accuracy                           0.55        20
   macro avg       0.52      0.50      0.46        20
weighted avg       0.53      0.55      0.49        20


────────────────────────────────────────────────────────────
  Classifier : Logistic Regression
────────────────────────────────────────────────────────────
  Weighted Precision : 0.1684
  Weighted Recall    : 0.4000
  Weighted F1-Score  : 0.2370

              precision    recall  f1-score   support

    Negative       0.00      0.00      0.00         6
```

```
PS C:\Dav assignment 2> & C:/Python314/python.exe "c:/Dav assignment 2/main.py"
============================================================
  SENTIMENT ANALYSIS — CLAUDE 4 (ANTHROPIC) TWEETS
============================================================

Dataset loaded: 100 tweets

Sentiment Distribution:
sentiment
neutral    40
positive   30
negative   30

Preprocessing complete.

Train/Test Split:
  Training set : 80 tweets (80%)
  Testing set  : 20 tweets  (20%)

TF-IDF vectorization complete.
  Vocabulary size : 500 features
  N-gram range    : (1, 2)

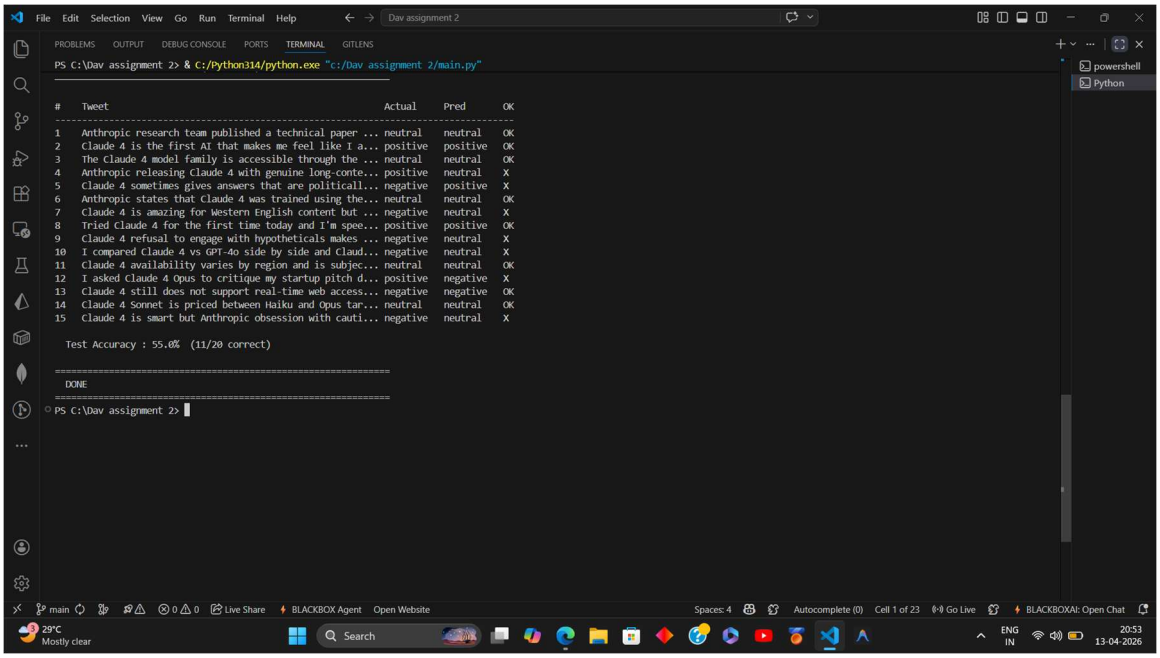
============================================================
  MODEL TRAINING & EVALUATION RESULTS
============================================================

────────────────────────────────────────────────────────────
  Classifier : Naive Bayes
────────────────────────────────────────────────────────────
  Weighted Precision : 0.4286
  Weighted Recall    : 0.5000
  Weighted F1-Score  : 0.4242

              precision    recall  f1-score   support

    Negative       0.33      0.17      0.22         6
     Neutral       0.57      1.00      0.73         8
    Positive       0.33      0.17      0.22         6

    accuracy                           0.50        20
```



```
PS C:\Dav assignment 2> & C:/Python314/python.exe "c:/Dav assignment 2/main.py"
================================================================

#   Tweet                                                  Actual     Pred       OK
----------------------------------------------------------------------------------------
1   Anthropic research team published a technical paper ... neutral    neutral    OK
2   Claude 4 is the first AI that makes me feel like I a... positive   positive   OK
3   The Claude 4 model family is accessible through the ... neutral    neutral    OK
4   Anthropic releasing Claude 4 with genuine long-conte... positive   neutral    X
5   Claude 4 sometimes gives answers that are politicall... negative   positive   X
6   Anthropic states that Claude 4 was trained using the... neutral    neutral    OK
7   Claude 4 is amazing for Western English content but ... negative   neutral    X
8   Tried Claude 4 for the first time today and I'm spee... positive   positive   OK
9   Claude 4 refusal to engage with hypotheticals makes ... negative   neutral    X
10  I compared Claude 4 vs GPT-4o side by side and Claud... negative   neutral    X
11  Claude 4 availability varies by region and is subjec... neutral    neutral    OK
12  I asked Claude 4 Opus to critique my startup pitch d... positive   negative   X
13  Claude 4 still does not support real-time web access... negative   negative   OK
14  Claude 4 Sonnet is priced between Haiku and Opus tar... neutral    neutral    OK
15  Claude 4 is smart but Anthropic obsession with cauti... negative   neutral    X

  Test Accuracy : 55.0%  (11/20 correct)

==========================================================
  DONE
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PS C:\Dav assignment 2>
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8. Conclusion & Future Scope

The project successfully demonstrates that even with a limited dataset of 100 samples, probabilistic models like Naïve Bayes can capture sentiment patterns with moderate accuracy. Future work involves scaling the dataset to 1,000+ entries and utilizing Transformer-based models like BERT for deeper contextual understanding.